FAST National University of Computer and Emerging Sciences
Islamabad Campus · Department of Software Engineering

Ref. No. _____
Date _____

REQUEST FOR ACCESS TO AGGREGATE BLOOD DEMAND DATA

Final Year Project · BS Software Engineering, 23rd Batch · Project title: BloodGrid

SECTION A · PURPOSE OF THIS REQUEST

This document formally requests access to historical, aggregate blood demand records held by your institution, for the sole purpose of academic research undertaken as a Final Year Project in the Department of Software Engineering, FAST National University of Computer and Emerging Sciences, Islamabad.

No patient-identifiable information, donor-identifiable information, or clinical record of any individual is requested or required. The request is limited to counts of blood units by date and blood group, together with a small number of operational indicators described in Section C.

Summary of what is being asked for

Daily or monthly counts of blood units issued, broken down by ABO and Rh blood group, covering the longest period your records allow. Where available, we also request indicators of days on which stock was unavailable, so that the analysis can distinguish blood that was requested from blood that was issued.

SECTION B · THE PROJECT, AND WHY THIS DATA IS NEEDED

B1. Project overview

BloodGrid is an academic software project addressing coordination in blood services. It combines two components: matching individual blood requests to suitable registered donors, and analysing historical blood demand in order to anticipate periods of elevated need before they occur.

B2. Why real data is necessary

The second component, blood demand analysis, is the subject of this request. A demand analysis model must be developed and evaluated against genuine historical demand. Synthetic or simulated data can demonstrate that a software system functions, but it cannot establish whether the analysis produces findings of any real value, because the patterns in simulated data are placed there by the person who generated it.

Access to authentic records would allow the project to determine whether demand for particular blood groups follows patterns that can be identified in advance, and whether such a system could give a blood service useful advance notice of periods of elevated demand. Without real data the project can only demonstrate feasibility of the software, not usefulness of the analysis.

B3. Intended benefit to the participating institution

Should your institution wish, the project team will provide, at no cost and with no obligation:

- A written summary of any demand patterns identified in your data, including seasonal and weekly variation by blood group.
- A copy of the final project report and its findings.
- A demonstration of the resulting software system.

The project team makes no claim that its output is validated for operational or clinical use, and no such use is proposed. The work is academic in nature.

B4. Scope limitation

This request relates to historical records only. It does not seek access to any live system, any database connection, any patient record, any donor register, or any operational integration with your institution's systems.

SECTION C · DATA REQUESTED

Please supply whatever subset of the following your records permit. Items marked **Essential** are the minimum needed for the analysis to proceed. Partial data is welcome; the project will state clearly what was and was not available.

C1. Essential fields

FIELD	DESCRIPTION	AVAILABLE
Date	Date of issue. Daily records are strongly preferred. Monthly totals are acceptable if daily records are not held.	☐
Blood group	ABO and Rh recorded separately, that is A+, A−, B+, B−, AB+, AB−, O+, O−.	☐
Units issued	Count of units issued per blood group per period.	☐
Period covered	The longest continuous period available. Three years or more is preferred; a shorter period is still useful.	☐

C2. Additional fields, important for accuracy

These allow the analysis to distinguish blood that was *needed* from blood that was *issued*. Without them, demand during shortage periods is systematically understated, which is precisely the period the analysis is intended to address.

FIELD	WHY IT IS REQUESTED	AVAILABLE
Days at zero stock	Identifies periods where demand exceeded what could be issued.	☐
Unfulfilled or referred-out requests	Together with units issued, allows actual demand to be reconstructed.	☐
Units cross-matched	Cross-match counts sit closer to clinical demand than issue counts.	☐
Units returned unused	Distinguishes units issued from units actually transfused.	☐
Units discarded or expired	Separates wastage from patient demand.	☐

C3. Optional context

FIELD	WHY IT IS USEFUL	AVAILABLE
Requesting department or ward	Separates scheduled surgical demand from emergency demand.	☐
Elective or emergency indicator	These two components behave differently over time.	☐
Changes in recording practice	Any change of system, capacity, or procedure that would create an apparent step in the records.	☐

C4. Acceptable formats

Comma-separated values, Microsoft Excel, or any standard database export. Consistent column naming across years is helpful but not necessary; the project team will undertake all cleaning and reconciliation. Printed or scanned records can also be accommodated if no electronic form exists, subject to volume.

SECTION D · UNDERTAKINGS ON DATA HANDLING AND CONFIDENTIALITY

The project team gives the following undertakings in respect of any data provided.

D1. Purpose limitation	The data will be used solely for the academic project described in Section B, and for no other purpose, project, or party.
D2. No personal data	No patient names, identifiers, record numbers, ages, diagnoses, addresses, or contact details are requested. No donor identifiers are requested. Should any such information be present in a supplied file, it will be deleted before analysis and its presence reported to your institution.
D3. Secure storage	Data will be held on password-protected devices belonging to the named team members and the supervising faculty member, and will not be uploaded to any public repository, shared drive, or third-party service.
D4. Aggregate reporting	Findings will be reported only in aggregate and statistical form. No individual record will be reproduced.
D5. Attribution	Your institution will be named in the project report only with written permission. If permission is withheld, the source will be described generically, for example as "a regional blood centre".
D6. Retention and disposal	Data will be retained only for the duration of the project and any subsequent examination process, and will be securely deleted thereafter, or on written request from your institution, whichever is earlier.
D7. Right of withdrawal	Your institution may withdraw consent at any time, in writing and without giving reason, upon which all supplied data will be deleted and confirmation provided.
D8. Review before publication	On request, your institution will be given the opportunity to review any material referring to its data before submission or publication.

SECTION E · INSTITUTION DETAILS

To be completed by the supplying institution.

Name of institution			
Address			
Type of facility	☐ Regional blood centre ☐ Hospital blood bank ☐ Standalone blood bank ☐ Other: _____		
Name of representative			
Designation			
Telephone		**Email**	
Data custodian, if different			

E1. Data availability, as assessed by the institution

Earliest record available		**Latest record available**	
Granularity held	☐ Daily ☐ Weekly ☐ Monthly ☐ Transaction level		
Format to be supplied	☐ CSV ☐ Excel ☐ Database export ☐ Printed or scanned		
Conditions or restrictions			

SECTION F · DECLARATION BY THE PROJECT TEAM

We, the undersigned, confirm that the data requested will be used only as described in this document, that the undertakings in Section D will be observed in full, and that no attempt will be made to identify any individual from the data supplied.

Abdullah Tariq
Project Team Member
Reg. No. _____ Date _____

Malik Usman
Project Team Member
Reg. No. _____ Date _____

Rizwan Saeed
Project Team Member
Reg. No. _____ Date _____

SECTION G · ENDORSEMENT BY THE DEPARTMENT

The undersigned confirm that this project is a recognised Final Year Project of the Department of Software Engineering, FAST National University of Computer and Emerging Sciences, Islamabad, that the request described above is made in the course of supervised academic work, and that the department supports this request.

Dr. Isma ul Hassan
Project Supervisor, Department of Software Engineering
Signature and date Official stamp

Dr. Atif Aftab Ahmad Jilani
Head of Department, Software Engineering
Signature and date Official stamp

SECTION H · AUTHORISATION BY THE SUPPLYING INSTITUTION

I confirm that I am authorised to release the data described in Section C on behalf of the institution named in Section E, that the data supplied contains no patient-identifiable or donor-identifiable information, and that I have read and accept the undertakings set out in Section D.

Data approved for release	☐ In full, as requested ☐ In part, as noted in Section E ☐ Declined
Institution may be named in the report	☐ Yes ☐ No, describe generically

[NAME OF AUTHORISED REPRESENTATIVE]
Designation _____
Signature and date

Official stamp of the institution

CONTACT FOR THIS REQUEST

Project team contact		**Telephone**	
Email		**Supervisor email**	

This document is a request for voluntary cooperation and creates no legal or contractual obligation on the part of the supplying institution. A copy of the departmental ethics approval and a formal data-sharing agreement can be provided on request.